

Anchor Drift No More: Hierarchical Consistency-Guided Prompt Distillation for Incomplete Multimodal Learning

Ruiting Dai
rtdai@uestc.edu.cn
University of Electronic Science and
Technology of China
Chengdu, China

Zesen Cai
zesen.cai@std.uestc.edu.cn
University of Electronic Science and
Technology of China
Chengdu, China

Lisi Mo
morris@uestc.edu.cn
University of Electronic Science and
Technology of China
Chengdu, China

Guiduo Duan
guiduo.duan@uestc.edu.cn
University of Electronic Science and
Technology of China
Chengdu, China

Keren Shi
skr@alu.uestc.edu.cn
University of Electronic Science and
Technology of China
Chengdu, China

Tao He*
tao.he01@hotmail.com
University of Electronic Science and
Technology of China
Chengdu, China

Abstract

Web-scale content is rich in modalities yet frequently incomplete due to device limits, transmission errors, or privacy controls, making learning with missing modalities a core challenge. Prior reconstruction and alignment strategies often fail to preserve a stable class geometry when inputs are partial, leading to *anchor drift*—a shift of class prototypes between complete and incomplete views that distorts the shared representation space and degrades generalization. We introduce **HiCoD** (**H**ierarchical **C**onsistency-Guided **P**rompt **D**istillation), which learns a robust, class-anchored semantic space. HiCoD combines: (1) a modality-aware semantic graph that restores cross-modal structure under partial observations; (2) dual-level anchoring that unifies large-language-model-derived global category prototypes with top-K local exemplars to balance cross-modal coherence and modality-specific detail; and (3) multi-level distillation that aligns unimodal features, fused embeddings, and prompt-completed signals within a single anchor space. Across CMU-MOSI, CMU-MOSEI, and additional benchmarks, HiCoD sets a new state of the art under both fixed-pattern and random missingness, improving Acc-2 by up to 6.4 points over MPLMM and remaining robust when key modalities are absent.

CCS Concepts

• **Computing methodologies** → **Learning latent representations**; *Supervised learning by classification*; • **Information systems** → Sentiment analysis.

Keywords

Incomplete Multimodal Learning; Anchor Drift; Prompt Distillation; Structure-aware Modelling; Class-Anchored Semantic Space

*Corresponding author.



This work is licensed under a Creative Commons Attribution 4.0 International License. WWW '26, Dubai, United Arab Emirates.

© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2307-0/2026/04
<https://doi.org/10.1145/3774904.3792498>

ACM Reference Format:

Ruiting Dai, Zesen Cai, Lisi Mo, Guiduo Duan, Keren Shi, and Tao He. 2026. Anchor Drift No More: Hierarchical Consistency-Guided Prompt Distillation for Incomplete Multimodal Learning. In *Proceedings of the ACM Web Conference 2026 (WWW '26)*, April 13–17, 2026, Dubai, United Arab Emirates. ACM, New York, NY, USA, 12 pages. <https://doi.org/10.1145/3774904.3792498>

1 Introduction

User-generated content (UGC) [15] underpins event understanding [25], and finance [17] across today's Web platforms. Yet, Web pipelines increasingly confront *incomplete multimodal signals*—videos without audio, images without captions, or privacy-redacted protection [20, 21]—stemming from bandwidth limits and consent controls. While multimodal learning has leveraged complementary signals to advance sentiment analysis [37], visual recognition [11, 13, 40, 40], visual tracking [45–47] and cross-modal retrieval [12, 44], missing modalities undermine standard fusion strategies that rely on cross-modal synergy.

Existing approaches to missing modalities largely fall into two families. *Explicit reconstruction* methods synthesize representations of the absent modality from observed ones (e.g., via autoencoders [8, 27], generative models [6, 42], or data selection [36]), aiming to restore input-level completeness. *Joint representation* methods avoid reconstruction and instead optimize shared or aligned embeddings (e.g., via contrastive [5] or co-regularized objectives [49]) to mitigate the impact of missing inputs.

Despite clear progress, both families face a structural limitation: when modalities are missing, the structure of the representation space can change in class-specific ways. We refer to this phenomenon as **anchor drift**—a systematic shift of class prototypes (centroids) [26] between the distributions of complete- vs. incomplete-modality inputs. Anchor drift arises from (i) the distributional gap between complete and incomplete instances and (ii) modality-specific variability that destabilizes the learned structure. As missing rates increase, this drift can induce boundary collapse and hinder generalization. Figure 1 visualizes fused multimodal features under varying missingness, comparing a strong prior method (KB-MMC [16]) and our approach; the anchor shift becomes more severe as missingness grows.

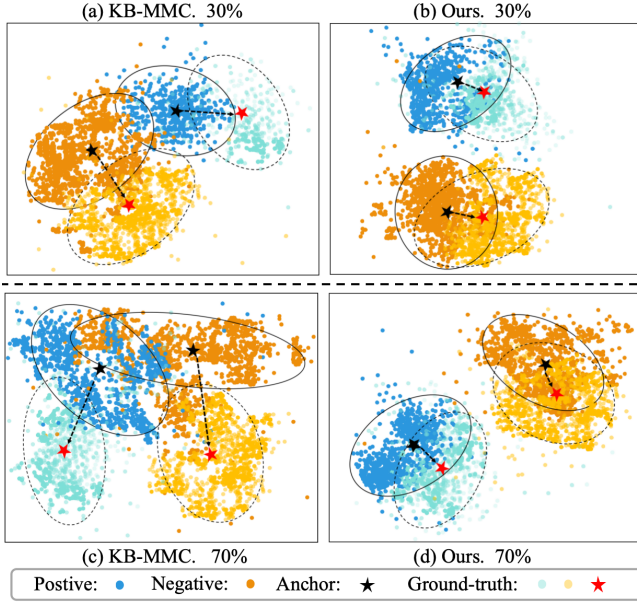


Figure 1: Visualization of binary emotion classification embeddings under different missing-modality ratios. As the missing-modality ratio increases from 30% to 70%, KB-MMC (a,c) shows anchor drift—missing modality prototypes (black star) shift from full modality centers (red star)—and boundary collapse. In contrast, HiCoD (ours) (b,d) preserves a class-anchored geometry with minimal drift. Points are samples (blue/orange = missing modality; light blue/light orange = complete modality); solid vs. dashed circles enclose missing-modality vs. full modalities distributions.

From a consistency-learning perspective, prior work primarily emphasizes (i) *intra-instance consistency* (aligning representations of the same sample across modalities)[4] and (ii) *inter-instance consistency* (matching same-class samples across modalities)[41]. In practice, intra-instance consistency may suppress modality-specific cues and over-penalize hard positives near dense boundaries (introducing false negatives), while inter-instance consistency often assumes cross-modal class congruence that breaks under unpaired or structurally inconsistent semantics. Figure 1 visualizes these limitations under high missing-rate conditions, highlighting prevalent anchor drift and boundary collapse—underscoring the inherent difficulty of balancing structural stability and semantic alignment in existing approaches. This raises a key question: *Can we move beyond reconstruction/alignment alone and learn representations that remain structurally stable as modalities drop out?*

Our answer is a three-level semantic consistency framework that: (1) enforces *local* (within-instance) modality consistency, (2) aggregates *inter-instance* semantics for fusion consistency, and (3) aligns all features in a shared *anchor space* for *global* distribution consistency. Under mild regularity assumptions on the encoders and anchor estimation, we provide a theoretical analysis showing that these constraints guide unimodal and fused features to converge toward common class-anchored directions.

Building on this insight, we propose **Hierarchical Consistency-Guided Prompt Distillation (HiCoD)** to construct a unified semantic space that is resilient to missing modalities. A structure-aware, modality-adaptive semantic graph stabilizes topology under partial modalities; dual-level (global category / local modality-specific) anchors unify semantic directions while preserving modality detail; and a multi-level distillation couples intra-modality transfer, cross-modal fusion, and semantic completion in the anchor space.

In summary, our contributions are as follows:

- **Analysis.** We systematically analyze performance degradation under missing modalities and argue that it principally arises from a lack of *semantic structural consistency*, manifested as anchor drift.
- **Method.** We introduce an anchor-driven, prompt-guided framework in which category-level anchors unify alignment across local features, fused representations, and high-level prompts, improving cross-modal coherence and robustness.
- **Modeling components.** We develop a modality-aware semantic graph with pseudo-label-driven adjacency and graph convolution to enhance structural stability under incomplete inputs, together with a unified multi-level distillation mechanism operating in the anchor space.
- **Results.** Extensive experiments across multiple benchmarks show consistent improvements under missing-modality settings, including robustness to different missing rates and key-modality ablations, with up to a ~ 6.4 point improvement in Acc-2 over the strong baseline MPLMM on CMU-MOSI and CMU-MOSEI.

2 Related Works

Generative Reconstruction. Early VAE and GAN methods follow a generate-then-fuse paradigm and reconstruct missing modalities via synthesis [3, 28], while diffusion-based models have recently shown strong recovery quality [2, 33]. Some approaches further incorporate external knowledge, such as knowledge graphs or large multimodal models, to enable training-free completion [16, 48]. Nevertheless, these methods still rely on a downstream fusion stage, where reconstruction errors can accumulate and induce semantic drift under high missing rates.

Cross-Modal Knowledge Distillation and Prompt Transfer. Beyond logit mimicry, modern KD transfers richer structural semantics (e.g., relations, prototypes, or hierarchical cues) from full-modality teachers to missing-modality students [22]. Prompt distillation leverages frozen LLMs together with retrieved or generated anchors to inject high-level semantics without reconstructing raw signals [18, 35]. By unifying KD and prompt engineering, recent methods leverage strong models or external knowledge to compensate for missing modalities with minimal computational cost [34].

Graph-Based Learning and Compensation. Graph neural networks compensate missing modalities via message passing over semantic, temporal, or speaker relations [23, 30] to hallucinate missing edges and recover neighborhood structure. Spectral regularization and adaptive edges improve robustness to fragmented inputs [7, 23]. Overall, graph learning models structured relations to provide contextual cues for missing modalities.

3 Preliminaries

Problem Statement. Consider a dataset with M modalities, where each sample ideally contains a complete set of features $X_i = (x_i^{(1)}, x_i^{(2)}, \dots, x_i^{(M)})$, with $x_i^{(m)} \in \mathbb{R}^{d_m}$ representing the feature vector of modality m . In practice, due to constraints such as transmission failures or privacy concerns, some modalities may be missing. Let $\mathcal{A}_i \subseteq \{1, \dots, M\}$ denote the set of available modalities for sample X_i . The valid input is thus $\{x_i^{(m)}\}_{m \in \mathcal{A}_i}$. Our goal is to learn a classification function

$$\mathcal{F} : \{x_i^{(m)} : m \in \mathcal{A}_i\} \rightarrow y_i, \quad y_i \in \{1, \dots, C\},$$

that enables robust multimodal classification under arbitrary combinations of available modalities.

Notations. We denote modality-specific embeddings by

$$z_i^{(m)} = \phi_m(x_i^{(m)}) \in \mathbb{R}^d,$$

and the fused embedding by

$$u_i = \Psi(\{x_i^{(m)}\}_{m \in \mathcal{A}_i}) \in \mathbb{R}^d.$$

Here, $\phi_m(\cdot)$ is the encoder for modality m , and $\Psi(\cdot)$ is the multimodal fusion function. Class centers (anchors) are collected in $C = [c_1, \dots, c_C] \in \mathbb{R}^{d \times C}$.

Definition 1 (Intra-instance Coherence). Different modalities of the same sample should remain consistent in a shared semantic space after encoding. For any two modalities $m, n \in \mathcal{A}_i$, their encoded representations should satisfy:

$$\|z_i^{(m)} - z_i^{(n)}\|_2 \leq \epsilon_1,$$

Definition 2 (Inter-instance Coherence). Samples from the same class should lie close in the fused representation space. For any two samples X_i and X_j from the same class, their fused embeddings should satisfy:

$$\|u_i - u_j\|_2 \leq \epsilon_2,$$

Definition 3 (Global semantic consistency). For any representation $z \in \mathbb{R}^d$ (either modality-specific $z_i^{(m)}$ or fused u_i), define the class-similarity distribution

$$s(z) = \text{softmax}(\frac{1}{\tau} C^\top z) \in \Delta^{C-1},$$

where $\Delta^{C-1} = \{p \in \mathbb{R}^C : p_c \geq 0, \sum_{c=1}^C p_c = 1\}$, and $\tau > 0$ is a temperature parameter. For any same-class pair z_p, z_q ,

$$\|s(z_p) - s(z_q)\|_1 \leq \epsilon,$$

i.e., same-class features produce similar anchor-aligned distributions, regardless of whether they are unimodal or fused. (See Appendix B for discussion and variants.)

Proposition 1 (Semantic Convergence). Assuming Definitions 1 and 2 hold, same-class representations tend to align in the anchor space. Specifically, for any two representations $z_p, z_q \in \{z_i^{(m)}, u_i\}$ from the same class, their anchor distributions (Def. 3) satisfy

$$\|s(z_p) - s(z_q)\|_1 \leq \epsilon,$$

As $\epsilon_1, \epsilon_2 \rightarrow 0$, we also have $\epsilon \rightarrow 0$, indicating that all same-class representations converge to a consistent direction in the anchor space. Full derivation is provided in Appendix A.

4 Methodology

Our framework (Fig. 2) comprises three key components: (1) Structure-aware representation modeling, which leverages multimodal pretrained encoders and a pseudo-label-based semantic graph with GCNs to capture high-level features and preserve topology under missing modalities; (2) LLM-guided dual-level semantic anchoring, where category prototypes from a large language model and Top-K confident samples form local and global anchors to enhance semantic discrimination and cross-modal alignment; and (3) Hierarchical consistency-guided prompt distillation, which enforces consistency at modality, fusion, and language levels to prevent semantic drift against missing modalities.

4.1 Structure-aware Representation Learning

Missing modalities often disrupt the relational structure between samples, leading to suboptimal fusion. To address this, we (i) project all modalities into a shared space, (ii) create a graph that captures both sample- and class-level relationships, and (iii) enhance the fused embedding by graph convolutional networks (GCNs).

Modality Encoding. We first extract modality-specific features using frozen, pre-trained encoders:

$$f_i^{(m)} = g_E^{(m)}(x_i^{(m)}), \quad f_i^{(m)} \in \mathbb{R}^{d_m},$$

where $x_i^{(m)}$ is the input of modality m , and $g_E^{(m)}$ is its encoder (e.g., ViT-B/16 for images, BERT-base for text, Wav2Vec 2.0 for audio). To align features across modalities, we apply lightweight adapters implemented with MLPs:

$$z_i^{(m)} = \text{Adapter}^{(m)}(f_i^{(m)}),$$

where $z_i^{(m)} \in \mathbb{R}^D$ is the projected embedding.

Fused Multimodal Representation. For each sample i , given its available modalities $\mathcal{A}_i \subseteq \{1, \dots, M\}$, the fused embedding is:

$$u_i = \sum_{m \in \mathcal{A}_i} \omega_i^{(m)} z_i^{(m)}, \quad \omega_i^{(m)} = \frac{1 + \lambda}{|\mathcal{A}_i| + \lambda M}, \quad (1)$$

Here M is the total number of modalities, and $\lambda > 0$ is a smoothing hyperparameter. When $\lambda = 0$, the fusion reduces to a simple average over the available modalities ($\omega_i^{(m)} = 1/|\mathcal{A}_i|$), ensuring equal contribution from each observed view. As $\lambda \rightarrow \infty$, the weights converge to $\omega_i^{(m)} \rightarrow 1/M$, so that $\sum_{m \in \mathcal{A}_i} \omega_i^{(m)} \rightarrow |\mathcal{A}_i|/M$. In this context, each modality is bounded by the uniform allocation $1/M$, which prevents a single observed modality from dominating the fused representation when others are missing. Thus λ smoothly interpolates between averaging of observed modalities and uniform weighting over all modalities, mitigating the effect of modality drop.

Cross-Modal Semantic Graph Construction. To model latent structure relations under missing modalities, we construct a heterogeneous graph that integrates sample-to-sample and sample-to-centers semantics. Specifically, the graph construction consists of the following three steps:

(1) *Pseudo-Label and Class Center Generation:* For each modality m , given a sample embedding $z_i^{(m)}$, we generate soft pseudo-labels by:

$$p_i^{(m)} = \text{softmax} \left(\frac{\text{Classifier}_m(z_i^{(m)})}{\tau} \right), \quad p_i^{(m)} \in \mathbb{R}^C, \quad (2)$$

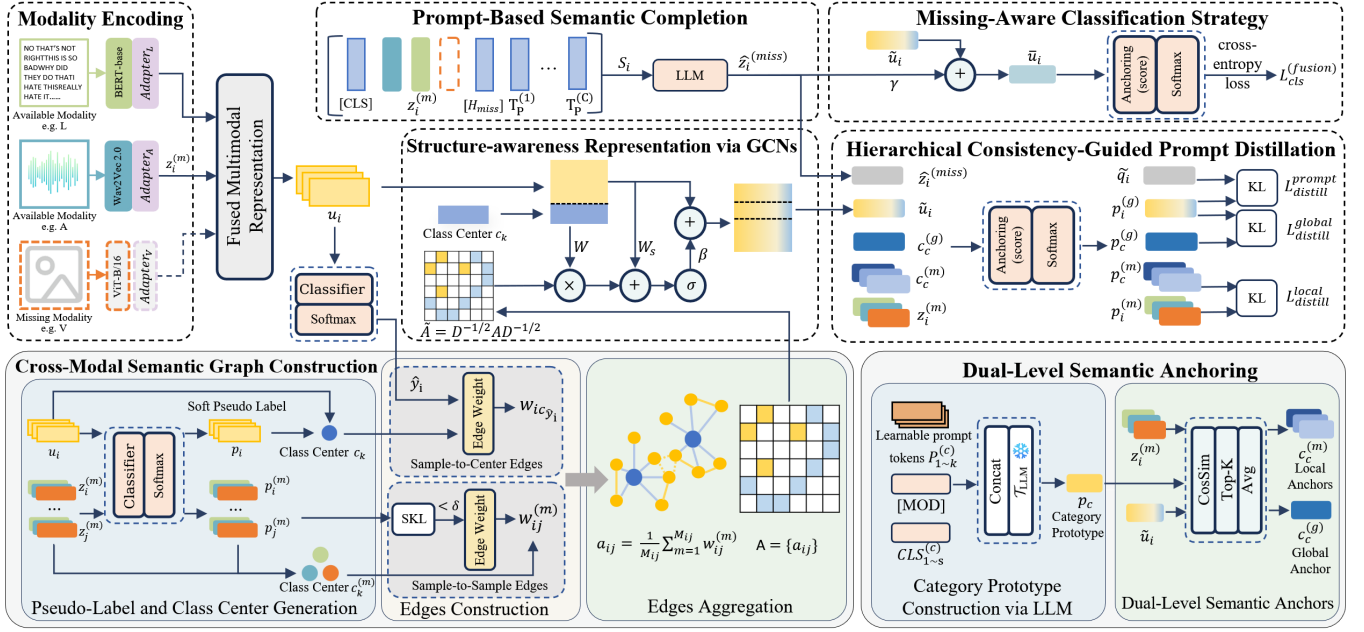


Figure 2: Overview of the proposed framework, which comprises structure-aware representation learning with pseudo-label-based semantic graphs, LLM-guided dual-level semantic anchoring, and hierarchical consistency-guided prompt distillation to enhance semantic consistency under missing modalities.

where $\tau > 0$ is a temperature parameter controlling prediction smoothness, and C is the number of classes. Here Classifier_m is a modality-specific linear layer, i.e., $\text{Classifier}_m(z) = W_m z + b_m$ with $W_m \in \mathbb{R}^{C \times D}$. The predicted pseudo-class is $\hat{y}_i^{(m)} = \arg \max(\mathbf{p}_i^{(m)})$. Then we compute class centers per modality as:

$$c_k^{(m)} = \frac{1}{|S_k^{(m)}|} \sum_{i \in S_k^{(m)}} z_i^{(m)}, \quad S_k^{(m)} = \{i \mid \hat{y}_i^{(m)} = k\}, \quad (3)$$

which serve as structural anchors to stabilize graph connectivity and compensate for pseudo-label fluctuations.

(2) *Sample-to-sample Edges Construction*: To compute the similarity between samples i and j , we define the symmetric KL divergence between their pseudo-labels:

$$\text{SKL}_{ij}^{(m)} = \frac{1}{2} [\text{KL}(p_i^{(m)} \| p_j^{(m)}) + \text{KL}(p_j^{(m)} \| p_i^{(m)})]. \quad (4)$$

An edge between sample nodes is established only if $\text{SKL}_{ij}^{(m)} < \delta$, with the edge weight defined as:

$$w_{ij}^{(m)} = \left(1 - \frac{\text{SKL}_{ij}^{(m)}}{\delta}\right) \cdot \exp\left(-\frac{1}{2} \|z_i^{(m)} - c_{\hat{y}_j^{(m)}}^{(m)}\|^2\right) \cdot \exp\left(-\frac{1}{2} \|z_j^{(m)} - c_{\hat{y}_i^{(m)}}^{(m)}\|^2\right). \quad (5)$$

The term $\exp(-\frac{1}{2} \|z_i^{(m)} - c_{\hat{y}_j^{(m)}}^{(m)}\|^2)$ measures, under modality m , how close sample i is to the class center of j 's predicted class. Similarly, $\exp(-\frac{1}{2} \|z_j^{(m)} - c_{\hat{y}_i^{(m)}}^{(m)}\|^2)$ measures how close sample j is to the class center of i 's predicted class.

(3) *Sample-to-Center Edges*: Besides, each sample node is also connected to its predicted class center node. Specifically, we apply a fused classifier to the fused representation u_i to obtain the distribution $\mathbf{p}_i$ as in Eq. (2), and then set $\hat{y}_i = \arg \max(\mathbf{p}_i)$. Class centers are then computed following Eq. (3) by averaging the fused representations u of all samples with predicted class k ; let $c_{\hat{y}_i}$ denote the corresponding center. The edge between sample i and $c_{\hat{y}_i}$ is assigned the following weight:

$$w_{ic_{\hat{y}_i}} = \exp\left(-\frac{1}{2} \|u_i - c_{\hat{y}_i}\|^2\right). \quad (6)$$

(4) *Edges Aggregation*: For each sample pair (i, j) , the edge weight is obtained by averaging over all shared modalities:

$$a_{ij} = \frac{1}{M_{ij}} \sum_{m=1}^{M_{ij}} w_{ij}^{(m)}, \quad (7)$$

where M_{ij} is the number of modalities in which both i and j appear. The resulting graph $\mathbf{A} \in \mathbb{R}^{(N+C) \times (N+C)}$ captures both sample- and class-level semantic structure.

Structure-awareness Representation via GCNs. We enhance the joint representation $U = [u_1, \dots, u_N, c_1, \dots, c_C]^T$ (samples u_i and class centers c_k) using a single-layer GCN over the unified graph $\mathbf{A}$:

$$s_i = \sigma(D^{-\frac{1}{2}} \mathbf{A} D^{-\frac{1}{2}} U W + U W_s)_i, \quad (8)$$

where D is the degree matrix, $W, W_s \in \mathbb{R}^{D \times D}$ are learnable weights of the GCN, $\sigma(\cdot)$ is the activation, and $(\cdot)_i$ denotes taking the i -th row after the affine transformation, corresponding to node i . Samples without valid modality predictions ($\mathcal{A}_i = \emptyset$) are set to

$s_i = \mathbf{0}$. The final structure-enhanced feature is

$$\tilde{u}_i = u_i + \beta s_i, \quad (9)$$

where $\beta > 0$ controls the residual strength.

This structure-aware module is essential because missing modalities distort the relational structure among samples, which cannot be corrected by simple weighted fusion of the available modalities. By explicitly modeling sample-to-sample and sample-to-class relations, the semantic graph restores connectivity and stabilizes representation learning. Together with GCN propagation, this neighborhood information sharing encourages intra-class consistency and helps maintain a stable topology, thereby improving robustness to both noise and missing modalities.

Robustness to Pseudo-Label Noise. Pseudo-labels may inevitably contain noise, especially under high missing-rate conditions, which can lead to error propagation in graph-based learning. Our design mitigates this risk in three ways. First, soft pseudo-labels with temperature scaling preserve uncertainty and reduce the chance that over-confident mistakes dominate graph construction [29]. Second, connecting samples to class-level centers provides stable anchors, so that individual noisy predictions have limited impact on the overall structure [9]. Third, we gate edge construction through symmetric distributional agreement (symmetric KL) and suppress low-confidence links, ensuring that unreliable connections are weakened before message passing [31]. Together, these mechanisms strengthen the semantic graph against pseudo-label noise and, more broadly, improve its robustness under both label noise and missing modalities.

4.2 Dual-Level Semantic Anchoring

While structure-aware learning captures data topology, the representations remain vulnerable to semantic drift under modality loss [14, 31]. Moreover, the learned embedding lack unified category-knowledge guidance for consistent alignment and struggle to balance modality-specific cues with cross-modal coherence. To address this, we introduce LLM-derived category prototypes as stable semantic anchors, leveraging the rich knowledge of large language models to guide representation learning even with incomplete modalities. Building on our Semantic Anchoring Proposition (**Prop. 1**), we design a dual-level anchoring technique: (i) local anchors from Top-K high-confidence samples capture modality-specific details for discrimination, while (ii) global anchors unify fused representations for cross-modal consistency. Local and global anchors play complementary roles: local anchors capture modality-specific discriminability, while global prototypes promote cross-modal semantic alignment.

Category Prototype Construction via LLM. For each category $c \in \{1, \dots, C\}$, we design a text prompt:

$$T_p^{(c)} = [P_1^{(c)} \dots P_k^{(c)}] \parallel [\text{MOD}] \parallel [\text{CLS}_1^{(c)} \dots \text{CLS}_s^{(c)}], \quad (10)$$

where $P_j^{(c)}$ are learnable tokens, [MOD] specifies the modality (e.g., “This is a video.”), and $\text{CLS}_j^{(c)}$ encodes class semantics. A frozen LLM encoder $\mathcal{T}_{\text{LLM}}$ (e.g., T5) maps the prompt to a category prototype: $p_c = \mathcal{T}_{\text{LLM}}(T_p^{(c)}) \in \mathbb{R}^D$.

Dual-Level Semantic Anchors. Following *Local Modality Consistency* (**Def. 1**) and *Global Aggregation* (**Def. 2**), we construct local

anchors $c_c^{(m)}$ from unimodal features and a global anchor $c_c^{(g)}$ from structure-enhanced fused embeddings. For each sample $z_i^{(m)}$, we compute its similarity to the category prototype p_c via

$$s_{i,c}^{(m)} = \frac{\langle z_i^{(m)}, p_c \rangle}{\|z_i^{(m)}\| \cdot \|p_c\|} + \lambda H(\alpha_i), \quad (11)$$

where $\lambda > 0$ balances the entropy term $H(\alpha_i)$. The Top-K samples $\mathcal{I}_c^{(m)} = \text{TopK}(\{s_{i,c}^{(m)}\}_{i=1}^N)$ are then used to form anchors: $c_c^{(m)} = \frac{1}{|\mathcal{I}_c^{(m)}|} \sum_{i \in \mathcal{I}_c^{(m)}} z_i^{(m)}$, where $c_c^{(m)}$ denotes the local anchor (per modality). Similarly, global anchors $c_c^{(g)}$ are derived from $\tilde{u}_i$ in Eq.(9) using the same top-K selection as $c_c^{(g)} = \frac{1}{|\mathcal{I}_c^{(g)}|} \sum_{i \in \mathcal{I}_c^{(g)}} \tilde{u}_i$.

4.3 Hierarchical Consistency-Guided Prompt Distillation

To enforce cross-modal consistency under missing modalities, we propose a category-anchor-based multi-level distillation framework with three complementary branches: (1) Local-Level Distillation, aligning unimodal features with local anchors to retain modality-specific semantics; (2) Global-Level Distillation, aligning fused features with global anchors for cross-modal consistency; and (3) Prompt-Based Completion, leveraging LLM-generated prompts to inject high-level semantics into incomplete samples. These branches jointly guide representations toward a unified, category-aligned space.

Local-Level Distillation For sample i under modality m , we denote its representation as $z_i^{(m)} \in \mathbb{R}^D$ and the category c local anchor as $c_c^{(m)} \in \mathbb{R}^D$. The classification distribution is computed via:

$$p_i^{(m)}(c) = \frac{\exp(\langle z_i^{(m)}, c_c^{(m)} \rangle / \tau)}{\sum_{c'=1}^C \exp(\langle z_i^{(m)}, c_{c'}^{(m)} \rangle / \tau)}, \quad (12)$$

where $\langle \cdot, \cdot \rangle$ denotes similarity and $\tau > 0$ is a temperature factor. The distillation loss is defined as:

$$\mathcal{L}_{\text{distill}}^{\text{local}} = \sum_{i=1}^N \sum_{m=1}^M \text{KL}(p_i^{(m)} \parallel \text{stop_grad}(p_c^{(m)})), \quad (13)$$

where $\text{KL}(\cdot \parallel \cdot)$ is the Kullback–Leibler divergence, and $\text{stop_grad}(\cdot)$ prevents backpropagation.

Global-Level Distillation. For the i -th sample, let $\tilde{u}_i \in \mathbb{R}^D$ be its fusion-enhanced feature and $c_c^{(g)} \in \mathbb{R}^D$ the global anchor of category c . The category probability is:

$$p_i^{(g)}(c) = \frac{\exp(\langle \tilde{u}_i, c_c^{(g)} \rangle / \tau)}{\sum_{c'=1}^C \exp(\langle \tilde{u}_i, c_{c'}^{(g)} \rangle / \tau)}, \quad (14)$$

where $\tau > 0$ is the temperature. The distillation loss is:

$$\mathcal{L}_{\text{distill}}^{\text{global}} = \sum_{i=1}^N \text{KL}(p_i^{(g)} \parallel \text{stop_grad}(p_c^{(g)})), \quad (15)$$

with $\text{KL}(\cdot \parallel \cdot)$ denoting Kullback–Leibler divergence and $\text{stop_grad}(\cdot)$ preventing gradient flow through the teacher. Notably, local distillation emphasizes intra-modal discriminative ability, while global distillation strengthens semantic aggregation and collaboration among modalities.

Prompt-Based Compensation Distillation. To compensate for missing modalities, we leverage LLM-based semantic completion.

For sample i with available modalities $\mathcal{A}_i \subseteq \{1, \dots, M\}$, we construct an input sequence:

$$\mathcal{S}_i = [\text{CLS}] \parallel \{z_i^{(m)}\}_{m \in \mathcal{A}_i} \parallel [H_{\text{miss}}] \parallel T_p^{(1)} \parallel \dots \parallel T_p^{(C)}, \quad (16)$$

where $[H_{\text{miss}}]$ is a placeholder and $T_p^{(c)}$ are category prompt templates. A frozen LLM encodes $\mathcal{S}_i$ to produce the missing-modality embedding $\hat{z}_i^{(\text{miss})} \in \mathbb{R}^D$, which is mapped to the global anchor space to compute:

$$\tilde{q}_i(c) = \frac{\exp(\langle \hat{z}_i^{(\text{miss})}, c_c^{(g)} \rangle / \tau)}{\sum_{c'} \exp(\langle \hat{z}_i^{(\text{miss})}, c_{c'}^{(g)} \rangle / \tau)}. \quad (17)$$

The distillation loss is:

$$\mathcal{L}_{\text{distill}}^{\text{prompt}} = \sum_i \text{KL}(\tilde{q}_i \parallel p_i^{(g)}), \quad (18)$$

applied only during training to simulate missing scenarios and enable semantic recovery.

Missing-Aware Classification Strategy. We introduce a missing flag $\delta_i \in \{0, 1\}$ (1 if sample i has missing modalities) and a completion weight $\gamma > 0$. The fusion compensation representation is:

$$\tilde{u}_i = (1 - \delta_i)\tilde{u}_i + \delta_i(\tilde{u}_i + \gamma \hat{z}_i^{(\text{miss})}), \quad (19)$$

where $\tilde{u}_i$ is always preserved regardless of δ_i , so that the representation consistently retains semantic cues from available modalities. When $\delta_i = 1$, the missing-modality embedding $\hat{z}_i^{(\text{miss})}$ is incorporated as an auxiliary signal to mitigate semantic drift under missing modalities. This design enables the model to adaptively incorporate the missing-modality embedding without discarding valid information from available modalities. The final category probability is computed as:

$$p_i^{(\text{final})}(c) = \frac{\exp(\langle \tilde{u}_i, c_c^{(g)} \rangle / \tau)}{\sum_{c'} \exp(\langle \tilde{u}_i, c_{c'}^{(g)} \rangle / \tau)}, \quad (20)$$

and the classification loss is:

$$\mathcal{L}_{\text{cls}}^{(\text{fusion})} = - \sum_{i=1}^N \log p_i^{(\text{final})}(y_i), \quad (21)$$

where y_i is the ground-truth label.

Overall Training Objective. The final training objective combines classification loss and multiple distillation losses:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{cls}}^{(\text{fusion})} + \sum_k \lambda_k \mathcal{L}_{\text{distill}}^k,$$

where $k \in \{\text{local}, \text{global}, \text{prompt}\}$ and $\lambda_k > 0$ are weighting hyperparameters.

5 Experiments

Datasets and Evaluation Metrics. We evaluate our method on two standard multimodal emotion recognition benchmarks: CMU-MOSI [38] and CMU-MOSEI [1]. Performance is measured by three metrics: binary accuracy (Acc-2), macro F1 score (F1), and 7-class accuracy (Acc-7), under various missing-modality conditions.

Baseline Methods. We compare our approach with 9 state-of-the-art methods for missing-modality scenarios: DCCAE [32], TATE [39], MissModal [24], MPLMM [10], MMIN [43], GCNet [23], IMDer [33], MoDiCF [19], and KB-MMC [16]. For MoDiCF, we replicate its modality diffusion recovery and generate counterfactual samples by masking and replacing dominant modalities, enabling robust emotion classification.

Implementation Details. We extract text features with **BERT-base** (768 d), visual expression features with **Facet** (35 d), and acoustic features with **COVAREP** (74 d). Each modality is projected into a shared 512 d space via a learned head. Category prompts are encoded by a **frozen T5-Base** text encoder with learnable context tokens (prompt template in Appendix C). We optimize with **AdamW** using a learning rate of 5×10^{-5} for BERT and 1×10^{-3} for all other components, batch size 32, and train for 20 epochs, selecting the checkpoint with the best validation F1. Unless noted, key scalars are: $k = 15$ and $\delta = 1.5$ in graph construction, temperature $\tau=0.7$, graph gate $\beta=0.5$, and loss weights $(\lambda_{\text{local}}, \lambda_{\text{global}}, \lambda_{\text{prompt}})=(1.0, 1.0, 0.5)$ with $\gamma=0.3$. For more details, please refer to Appendix C.

5.1 Comparison with state-of-the-art methods

We compare our approach to strong reconstruction and joint-representation baselines under *fixed* and *random* missing-modality protocols (Tab. 1 and Fig. 3). Unless noted, tables report mean over five seeds. Moreover, we have provided more experimental results under different missing rates in Appendix D.

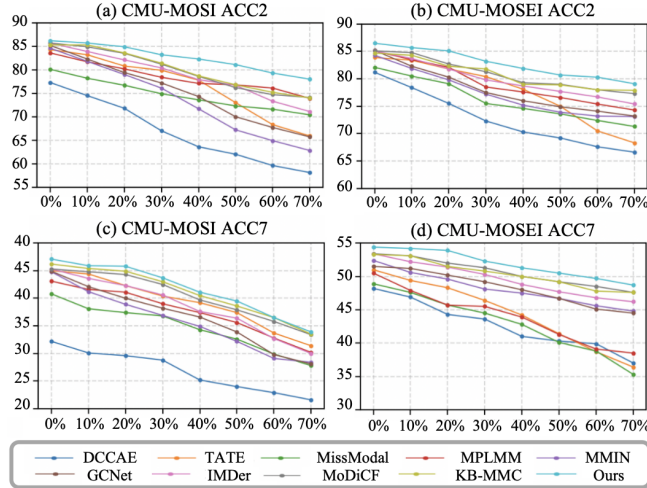
Overall accuracy across fixed-missing configurations. Across CMU-MOSI and CMU-MOSEI, our method achieves best or statistically tied results in the majority of fixed-missing settings (Table 1). Averaged over datasets and configurations, we improve *F1* and *Acc-2* by approximately 4–5 pp over joint-learning baselines (e.g., DCCAE, TATE, MissModal), and by 2–3 pp over leading reconstruction methods (e.g., IMDer, DiCMoR). These gains suggest that the proposed graph-structured fusion with anchor-guided distillation is more effective than directly fusing incomplete modalities. See the last row of Table 1 for relative improvements.

Robustness when key modalities are absent. Text is typically the most informative modality for sentiment; most methods degrade sharply when it is removed. In the {A,V} fixed-missing setting, our model remains leading on both datasets (Table 1), with margins that are positive and often significant. We attribute this to the multi-level anchor distillation, which preserves class-consistent structure in remaining modalities and reduces reliance on a single modality.

Resilience across increasing missing rates. Under random missingness, our model degrades gracefully: performance declines smoothly as the missing-modality ratio (r_{miss}) increases from (0%) to (70%) (Fig. 3). On CMU-MOSEI, the total reduction in *Acc-2* stays within (8) percentage points over this full range, indicating strong robustness compared to baselines whose curves become markedly steeper in the high-missingness regime. The near-monotonic, gently sloped trend suggests that graph propagation reliably transfers complementary cues from observed modalities and neighboring instances, while anchor supervision keeps class prototypes stable in the embedding space. Together, these

Table 1: Performance on CMU-MOSI and CMU-MOSEI in terms of Acc-2 and F1 under various missing modalities (L, A, V, and their combinations) with a 70% missing rate. +x indicates improvement over the second-best. Bold denotes the best.

	Model	{L}	{A}	{V}	{L,A}	{L,V}	{A,V}	{L,A,V}
CMU-MOSI	DCCA [32]	76.4 / 76.5	48.8 / 42.1	52.6 / 51.1	77.0 / 77.0	76.7 / 76.8	54.0 / 52.5	76.4 / 75.9
	TATE [39]	85.7 / 85.6	59.4 / 55.8	60.9 / 58.0	85.8 / 85.4	86.1 / 85.7	64.2 / 62.9	85.1 / 85.2
	MissModal [24]	79.1 / 79.2	56.1 / 54.5	55.0 / 54.4	81.0 / 81.0	81.1 / 81.2	57.5 / 57.4	80.8 / 80.2
	MPLMM [10]	80.1 / 80.3	62.7 / 63.7	63.1 / 63.7	80.8 / 81.1	81.1 / 81.2	65.0 / 65.4	81.6 / 81.5
	MMIN [43]	83.8 / 83.8	55.3 / 51.5	57.0 / 54.0	84.0 / 84.0	83.8 / 83.9	60.4 / 58.5	84.6 / 84.4
	GCNet [23]	83.7 / 83.6	56.1 / 54.5	56.1 / 55.7	84.5 / 84.4	84.3 / 84.2	62.0 / 61.9	85.2 / 85.1
	IMDer [33]	84.8 / 84.7	62.0 / 62.2	61.3 / 60.8	85.4 / 85.3	85.5 / 85.4	63.6 / 63.4	85.7 / 85.6
	MoDiCF [19]	84.6 / 84.6	60.7 / 61.5	62.1 / 61.2	85.5 / 85.4	85.5 / 85.6	64.4 / 63.7	85.7 / 85.6
	KB-MMC [16]	85.1 / 84.7	62.6 / 63.8	62.3 / 61.2	85.5 / 85.6	85.6 / 85.8	64.9 / 64.5	85.6 / 85.5
	Ours	87.5 +1.8 / 87.3 +1.7	69.1 +6.4 / 69.0 +5.2	67.2 +4.1 / 67.2 +3.5	87.6 +1.8 / 87.4 +1.8	87.6 +1.5 / 87.5 +1.7	70.3 +5.3 / 70.2 +4.8	87.8 +2.1 / 87.6 +2.0
CMU-MOSEI	DCCA [32]	79.7 / 79.5	61.4 / 53.8	61.1 / 57.2	80.0 / 80.0	80.4 / 80.4	62.7 / 59.2	80.5 / 80.4
	TATE [39]	82.8 / 82.5	65.5 / 65.2	62.5 / 61.2	84.1 / 84.2	83.9 / 83.8	64.1 / 62.8	84.9 / 84.4
	MissModal [24]	82.6 / 82.8	62.7 / 54.5	62.6 / 57.1	83.5 / 83.3	83.2 / 83.2	63.7 / 62.7	83.1 / 83.0
	MPLMM [10]	79.1 / 79.2	67.3 / 68.7	67.3 / 69.4	80.5 / 80.4	80.1 / 80.1	68.2 / 69.9	81.1 / 81.4
	MMIN [43]	82.3 / 82.4	58.9 / 59.5	59.3 / 60.0	83.7 / 83.7	83.8 / 83.4	63.5 / 61.9	84.3 / 84.2
	GCNet [23]	83.0 / 83.2	60.2 / 60.3	61.9 / 61.6	84.3 / 84.4	84.3 / 84.4	64.1 / 57.2	85.2 / 85.1
	IMDer [33]	84.5 / 84.5	63.8 / 60.6	63.9 / 63.6	85.1 / 85.1	85.0 / 85.0	64.9 / 63.5	85.1 / 85.1
	MoDiCF [19]	84.3 / 84.4	63.2 / 61.4	63.6 / 63.7	84.9 / 85.0	84.8 / 84.9	65.5 / 64.7	85.0 / 85.1
	KB-MMC [16]	84.4 / 84.3	63.7 / 62.5	63.8 / 63.8	84.8 / 84.7	84.9 / 84.8	66.5 / 66.3	84.8 / 84.5
	Ours	88.5 +4.0 / 88.3 +3.8	69.8 +2.5 / 69.6 +0.9	70.2 +2.9 / 70.0 +0.6	88.6 +3.5 / 88.5 +3.4	88.4 +3.4 / 88.3 +3.3	71.2 +3.0 / 71.0 +1.1	88.5 +3.3 / 87.9 +2.8

**Figure 3: Comparison of Method Performance under Random Missing Rates (0%–70%) on CMU-MOSI and CMU-MOSEI (Acc-2 and Acc-7).**

components limit semantic drift, preserve decision boundaries, and maintain predictive reliability even when a large fraction of modalities is absent.

5.2 Ablation Studies

We evaluate the contribution of each component via controlled ablations covering module configuration, graph construction, anchor design, and sensitivity. Unless noted, all results use a fixed missing rate of $r_{\text{miss}}=70\%$, identical training settings across variants, and report mean over five seeds. Additional ablation results, such

Table 2: Comparison of different module configurations based on average Acc-2, F1, and Acc-7 scores across seven missing-modality scenarios on CMU-MOSI and CMU-MOSEI. Results are reported under a fixed missing rate of $r_{\text{miss}}=70\%$.

Variant	MOSI-Fixed	MOSEI-Fixed
Full (Ours)	79.6 / 79.5 / 38.3	80.7 / 80.5 / 55.2
w/o GCN	77.3 / 77.4 / 36.8	78.2 / 78.2 / 52.8
w/o Distill	68.6 / 67.2 / 30.8	70.2 / 68.0 / 44.8
w/o Anchor-distill	73.8 / 74.5 / 33.7	75.1 / 75.7 / 48.7
w/o Compensation-distill	78.3 / 78.0 / 37.1	79.2 / 78.9 / 53.7

as broader missing rates and extended variants, are provided in Appendix E for comprehensive analysis.

Module configuration. Table 2 compares five variants: *Full*, *w/o GCN* (remove structure-aware propagation; see Section *Structure-aware Representation Learning*), *w/o Distill*. (drop all distillation terms, keeping only $\mathcal{L}_{\text{sup}}$), *w/o Anchor-distill* (remove anchor teachers and $\mathcal{L}_{\text{local}}$), and *w/o Compensation-distill* (remove $\mathcal{L}_{\text{prompt}}$). All degradations relative to *Full* are non-trivial, confirming the necessity of each component. Removing GCN reduces performance by ≈ 2.3 pp (MOSI) and ≈ 2.5 pp (MOSEI) on *Acc-2*, underscoring the value of structure-aware modeling. When all distillation terms are disabled, the largest drop is observed, proving that distillation improves both semantic alignment and cross-modal transfer. Disabling anchor distillation results in a significant decrease, especially on F_1 , indicating that anchor-based supervision is central to multi-level alignment. Eliminating compensation distillation yields a moderate but consistent decrease, suggesting its utility for recovering semantics under severe missingness.

Table 3: Comparison of different graph construction strategies based on average Acc-2, F1, and Acc-7 scores across seven missing-modality scenarios on CMU-MOSI and CMU-MOSEI. All experiments use a fixed missing rate of $r_{\text{miss}}=70\%$.

Graph Strategy	MOSI-Fixed	MOSEI-Fixed
Ours	79.6 / 79.5 / 38.3	80.7 / 80.5 / 55.2
GT-Label	79.8 / 79.7 / 36.9	80.2 / 80.1 / 53.4

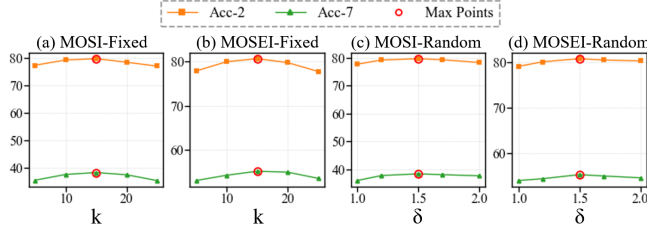


Figure 4: Performance under different parameters, i.e., k (Top-K for anchor selection) and δ (SKL threshold for edge gating) on CMU-MOSI and CMU-MOSEI. Results are reported under a fixed missing rate of $r_{\text{miss}}=70\%$.

Graph construction strategy. We compare (i) our *Pseudo-label* graph (δ -thresholded over soft class posteriors using symmetric KL divergence) to (ii) an *Oracle* (*GT-Labels*) graph that uses ground-truth labels *only to build edges* (privileged; analysis only). As shown in Table 3, pseudo-label graphs close much of the gap to the oracle and often match or exceed it on *Acc-7*, while remaining practical. We attribute this to preserving soft class distributions that capture semantic continuity and ambiguity—beneficial when modalities are missing—whereas GT labels enforce only hard intra-class ties.

Anchor construction strategy. Table 4 evaluates three strategies: *Top-K + LLM* (ours), *Top-K only* (no text guidance), and *Mean pooling*. The full strategy achieves the best results across *Acc-2*, *F1*, and *Acc-7*; relative to *Top-K only* it improves by ~ 1.4 pp on *Acc-2/F1* (often significant[†]), indicating that LLM-guided prompts produce more salient, discriminative anchors. The gap widens against mean pooling, affirming the importance of high-quality anchors for cross-modal alignment and distillation under missingness.

Hyperparameter sensitivity. Furthermore, we perform a sensitivity analysis on two key hyperparameters: the k (Top-K) used in anchor selection and the SKL threshold δ . As illustrated in Fig. 4, our method maintains consistently strong performance across a broad range of values for both parameters, demonstrating its robustness and practical viability.

Effect of anchor distillation on representation geometry. We visualize fused features with t-SNE under *Local-only*, *Global-only*, and *Dual-anchor* distillation (Fig. 5); all runs use the same seed and t-SNE hyperparameters. Local anchors tighten *intra-class* clusters (fine-grained semantics), global anchors enlarge *inter-class* margins, and the dual-anchor setting balances both, yielding the most separable structure. A k -NN probe on the embeddings corroborates the qualitative trend.

Table 4: Comparison of different anchor construction strategies based on average Acc-2, F1, and Acc-7 scores across seven missing-modality scenarios on CMU-MOSI and CMU-MOSEI. All results use a fixed missing rate of $r_{\text{miss}}=70\%$.

Anchor Strategy	MOSI-Fixed	MOSEI-Fixed
Top-K + LLM (our)	79.6 / 79.5 / 38.3	80.7 / 80.5 / 55.2
Top-K Only	78.2 / 78.1 / 37.2	79.3 / 79.2 / 53.5
Mean Pooling	76.0 / 75.7 / 34.8	77.0 / 76.6 / 50.0

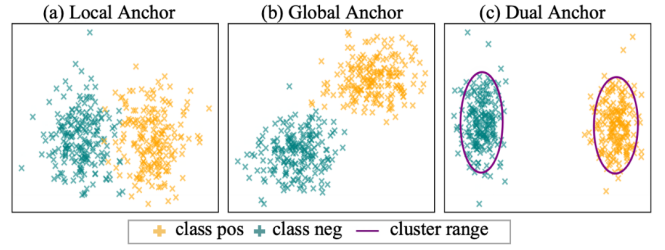


Figure 5: Comparison of Semantic Representations under Different Anchor Distillation Strategies on CMU-MOSEI via t-SNE Visualization.

6 Conclusion

We studied anchor drift in multimodal learning under missing modalities and showed why existing consistency objectives struggle to balance modality-specific detail with semantic alignment. We introduced a hierarchical consistency-guided prompt distillation framework built on three levels of semantic consistency. The approach combines pseudo-label-guided semantic graphs, prompt-driven dual-level anchors, and multi-channel distillation within a unified anchor space, yielding robust cross-modal representations. Extensive experiments across multiple benchmark datasets (e.g., CMU-MOSI, CMU-MOSEI) and diverse missingness protocols (both fixed-pattern and random modality dropout) demonstrate that HiCoD consistently outperforms state-of-the-art baselines.

Limitation and future work. Our framework introduces extra compute and design complexity: constructing semantic graphs (with pseudo-labels and SKL gating) and encoding text prompts add overhead and depend on calibration of the teacher distributions. Future work will explore more efficient graph construction (e.g., learned sparsification), prompt-free or self-supervised anchors that reduce reliance on label text, backbone adaptation under missingness, and evaluation under more extreme and open-set conditions.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (NSFC) (granted No. 62306064) and the Central-Guided Local Science and Technology Development (granted No. 2023ZYD0165). We appreciate all the authors for their fruitful discussions. In addition, thanks are extended to anonymous reviewers for their insightful comments.

References

- [1] AmirAli Bagher Zadeh, Paul Pu Liang, Soujanya Poria, Erik Cambria, and Louis-Philippe Morency. 2018. Multimodal Language Analysis in the Wild: CMU-MOSEI Dataset and Interpretable Dynamic Fusion Graph. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (ACL 2018), Volume 1: Long Papers*. 2236–2246. doi:10.18653/v1/P18-1208
- [2] Mustapha Bounoua, Giulio Franzese, and Pietro Michiardi. 2024. Multi-Modal Latent Diffusion. *Entropy* 26, 4 (2024). doi:10.3390/e26040320
- [3] Bing Cao, Zhiwei Bi, Qinghua Hu, Han Zhang, Nannan Wang, Xinbo Gao, and Dinggang Shen. 2023. AutoEncoder-Driven Multimodal Collaborative Learning for Medical Image Synthesis. *International Journal of Computer Vision* 131, 8 (2023), 1995–2014. doi:10.1007/s11263-023-01791-0
- [4] Chuanguan Chen, Zhencheng Li, Kit Ian Kou, Jie Du, Chen Li, Hongtao Wang, and Chi-Man Vong. 2025. Comprehensive Multisource Learning Network for Cross-Subject Multimodal Emotion Recognition. *IEEE Trans. Emerg. Topics Comput. Intell.* 9, 1 (2025), 365–380. doi:10.1109/TETCI.2024.3406422
- [5] Giordano Cicchetti, Eleonora Grassucci, Luigi Sigillo, and Danilo Comminiello. 2025. Gramian multimodal representation learning and alignment. In *Proc. ICLR 2025*. Singapore.
- [6] Ruiting Dai, Chenxi Li, Yandong Yan, Lisi Mo, Ke Qin, and Tao He. 2025. Unbiased Missing-modality Multimodal Learning. In *Proc. ICCV*.
- [7] Fangze Fu, Wei Ai, Fan Yang, Yuntao Shou, Tao Meng, and Keqin Li. 2025. SDR-GNN: Spectral Domain Reconstruction Graph Neural Network for incomplete multimodal learning in conversational emotion recognition. *Knowledge-Based Systems* 309 (2025), 112825. doi:10.1016/j.knsys.2024.112825
- [8] Pengxuan Gao, Tianyu Liu, Jia-Wen Liu, Bao-Liang Lu, and Wei-Long Zheng. 2024. Multimodal Multi-View Spectral-Spatial-Temporal Masked Autoencoder for Self-Supervised Emotion Recognition. In *Proc. ICASSP 2024*. 1926–1930. doi:10.1109/ICASSP48485.2024.10447194
- [9] Yanqi Ge, Qiang Nie, Ye Huang, Yong Liu, Chengjie Wang, Feng Zheng, Wen Li, and Lixin Duan. 2024. Beyond Prototypes: Semantic Anchor Regularization for Better Representation Learning. *AAAI* 38, 3 (Mar. 2024), 1887–1895. doi:10.1609/aaai.v38i3.27958
- [10] Zirun Guo, Tao Jin, and Zhou Zhao. 2024. Multimodal Prompt Learning with Missing Modalities for Sentiment Analysis and Emotion Recognition. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL 2024), Long Papers*. 1726–1736. doi:10.18653/v1/2024.acl-long.94
- [11] Tao He, Lianli Gao, Jingkuan Song, Jianfei Cai, and Yuan-Fang Li. 2021. Semantic compositional learning for low-shot scene graph generation. *arXiv preprint arXiv:2108.08600* (2021).
- [12] Tao He, Lianli Gao, Jingkuan Song, and Yuan-Fang Li. 2021. Semisupervised network embedding with differentiable deep quantization. *IEEE Transactions on Neural Networks and Learning Systems* 34, 8 (2021), 4791–4802.
- [13] Xin Hu, Ke Qin, Guiduo Duan, Ming Li, Yuan-Fang Li, and Tao He. 2025. SPADE: Spatial-Aware Denoising Network for Open-vocabulary Panoptic Scene Graph Generation with Long- and Local-range Context Reasoning. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 15562–15572.
- [14] Zhangqi Jiang, Tingjin Luo, and Xinyan Liang. 2024. Deep Incomplete Multi-View Learning Network with Insufficient Label Information. In *Proc. AAAI 2024*, Vol. 38. 12919–12927. doi:10.1609/aaai.v38i11.29189
- [15] Pranav Kasela, Marco Braga, Gabriella Pasi, and Raffaele Perego. 2024. SE-PQA: Personalized Community Question Answering. In *Companion Proceedings of The Web Conference 2024 (WWW '24)*. 1095–1098. doi:10.1145/3589335.3651445
- [16] Guanzhou Ke, Shengfeng He, Xiaoli Wang, Bo Wang, Guoqing Chao, Yuanyang Zhang, Yi Xie, and Hexing Su. 2025. Knowledge Bridger: Towards Training-Free Missing Modality Completion. In *Proc. CVPR*. 25864–25873.
- [17] Zong Ke, Yuqing Cao, Zhenrui Chen, Yuchen Yin, Shouchao He, and Yu Cheng. 2025. Early warning of cryptocurrency reversal risks via multi-source data. *Finance Research Letters* (2025), 107890.
- [18] Jian Lang, Zhangtao Cheng, Ting Zhong, and Fan Zhou. 2025. Retrieval-Augmented Dynamic Prompt Tuning for Incomplete Multimodal Learning. In *Proc. AAAI*. 18035–18043. doi:10.1609/aaai.v39i17.33984
- [19] Jin Li, Shoujin Wang, Qi Zhang, Shui Yu, and Fang Chen. 2025. Generating with fairness: A modality-diffused counterfactual framework for incomplete multimodal recommendations. In *Proceedings of the ACM on Web Conference 2025*. 2787–2798.
- [20] Ming Li, Huazhu Fu, Shengfeng He, Hehe Fan, Jun Liu, Jussi Keppo, and Mike Zheng Shou. 2023. DR-FER: Discriminative and Robust Representation Learning for Facial Expression Recognition. *IEEE Transactions on Multimedia* 26 (2023), 6297–6309.
- [21] Ming Li, Xiangyu Xu, Hehe Fan, Pan Zhou, Jun Liu, Jia-Wei Liu, Jiahe Li, Jussi Keppo, Mike Zheng Shou, and Shuicheng Yan. 2023. STPrivacy: Spatio-temporal privacy-preserving action recognition. In *ICCV*.
- [22] Mingcheng Li, Dingkan Yang, Xiao Zhao, Shuaibing Wang, Yan Wang, Kun Yang, Mingyang Sun, Dongliang Kou, Ziyun Qian, and Lihua Zhang. 2024. Correlation-Decoupled Knowledge Distillation for Multimodal Sentiment Analysis with Incomplete Modalities. In *Proc. CVPR 2025*. 12458–12468. doi:10.1109/CVPR52733.2024.01184
- [23] Zheng Lian, Lan Chen, Licai Sun, Bin Liu, and Jianhua Tao. 2023. GCNet: Graph Completion Network for Incomplete Multimodal Learning in Conversation. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 45, 7 (2023), 8419–8432. doi:10.1109/TPAMI.2023.3234553
- [24] Ronghao Lin and Haifeng Hu. 2023. MissModal: Increasing Robustness to Missing Modality in Multimodal Sentiment Analysis. *Transactions of the Association for Computational Linguistics* 11 (2023), 1686–1702. doi:10.1162/tacl_a_00628
- [25] Shaoyu Liu, Jianing Li, Guanghui Zhao, Yunjian Zhang, Xin Meng, Fei Richard Yu, Xiangyang Ji, and Ming Li. 2025. EventGPT: Event Stream Understanding with Multimodal Large Language Models. In *Proceedings of the Computer Vision and Pattern Recognition Conference (CVPR)*. 29139–29149.
- [26] Suyuan Liu, Siwei Wang, Ke Liang, Junpu Zhang, Zhibin Dong, Tianrui Liu, En Zhu, Kunlun He, and Xinwang Liu. 2024. Alleviate Anchor-Shift: Explore Blind Spots with Cross-View Reconstruction for Incomplete Multi-View Clustering. In *Advances in Neural Information Processing Systems*, Vol. 37. 87509–87531. doi:10.52202/079017-2777
- [27] Yao Lu, Ji Zhaiyuan, Jiawei Du, Yu Shanqing, Qi Xuan, and Joey Tianyi Zhou. [n. d.]. AutoAnnotator: A Collaborative Annotation Framework for Large and Small Language Models. *Transactions on Machine Learning Research* [n. d.].
- [28] Yixin Luo and Zhouwang Yang. 2024. DynGAN: Solving Mode Collapse in GANs With Dynamic Clustering. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 46, 8 (2024), 5493–5503. doi:10.1109/TPAMI.2024.3367532
- [29] Rafael Müller, Simon Kornblith, and Geoffrey E Hinton. 2019. When does label smoothing help?. In *Advances in Neural Information Processing Systems*, H. Wallach, H. Larochelle, A. Beygelzimer, F. d'Alché-Buc, E. Fox, and R. Garnett (Eds.), Vol. 32. Curran Associates, Inc.
- [30] Thu Hang Phung, Binh P. Nguyen, Thanh Hung Nguyen, Quoc Viet Hung Nguyen, Phi Le Nguyen, and Thanh Trung Huynh. 2024. A Contrastive Learning and Graph-based Approach for Missing Modalities in Multimodal Federated Learning. In *2024 International Joint Conference on Neural Networks (IJCNN)*. 1–8. doi:10.1109/IJCNN60899.2024.10650285
- [31] Botao Wang, Jia Li, Yang Liu, Jiashun Cheng, Yu Rong, Wenjia Wang, and Fuguee Tsung. 2023. Deep Insights into Noisy Pseudo Labeling on Graph Data. In *Advances in Neural Information Processing Systems*, A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine (Eds.), Vol. 36. Curran Associates, Inc., 76214–76228.
- [32] Weiran Wang, Raman Arora, Karen Livescu, and Jeff Bilmes. 2015. On Deep Multi-View Representation Learning. In *Proceedings of the 32nd International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 37)*. Lille, France, 1083–1092. <https://proceedings.mlr.press/v37/wangb15.html>
- [33] Yuanzhi Wang, Yong Li, and Zhen Cui. 2023. Incomplete Multimodality-Diffused Emotion Recognition. In *Advances in Neural Information Processing Systems*, A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine (Eds.), Vol. 36. Curran Associates, Inc., 17117–17128.
- [34] Wei Wei, Jiabin Tang, Lianghao Xia, Yangqin Jiang, and Chao Huang. 2024. PromptMM: Multi-Modal Knowledge Distillation for Recommendation with Prompt-Tuning. In *Proceedings of the ACM Web Conference 2024 (WWW '24)*. 3217–3228. doi:10.1145/3589334.3645359
- [35] Yingxue Xu, Fengtao Zhou, Chenyu Zhao, Yihui Wang, Can Yang, and Hao Chen. 2025. Distilled Prompt Learning for Incomplete Multimodal Survival Prediction. In *Proc. CVPR*. 5102–5111.
- [36] Jiawei Yao, Chuming Li, and Canran Xiao. 2024. Swift sampler: Efficient learning of sampler by 10 parameters. *Advances in Neural Information Processing Systems* 37 (2024), 59030–59053.
- [37] Wen Yin, Yong Wang, Guiduo Duan, Dongyang Zhang, Xin Hu, Yuan-Fang Li, and Tao He. 2025. Knowledge-Aligned Counterfactual-Enhancement Diffusion Perception for Unsupervised Cross-Domain Visual Emotion Recognition. In *Proc. CVPR*. 3888–3898.
- [38] Amir Zadeh, Rowan Zellers, Eli Pincus, and Louis-Philippe Morency. 2016. Multimodal Sentiment Intensity Analysis in Videos: Facial Gestures and Verbal Messages. *IEEE Intelligent Systems* 31, 6 (2016), 82–88. doi:10.1109/MIS.2016.94
- [39] Jiandian Zeng, Tianyi Liu, and Jiantao Zhou. 2022. Tag-assisted Multimodal Sentiment Analysis under Uncertain Missing Modalities. In *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval (Madrid, Spain) (SIGIR '22)*. 1545–1554. doi:10.1145/3477495.3532064
- [40] Shuang Zeng, Xinyuan Chang, Mengwei Xie, Xinran Liu, Yifan Bai, Zheng Pan, Mu Xu, and Xing Wei. 2025. FutureSightDrive: Thinking Visually with Spatio-Temporal CoT for Autonomous Driving. *arXiv preprint arXiv:2505.17685* (2025).
- [41] Xiaoheng Zhang, Weigang Cui, Bin Hu, and Yang Li. 2024. A Multi-Level Alignment and Cross-Modal Unified Semantic Graph Refinement Network for Conversational Emotion Recognition. *IEEE Transactions on Affective Computing* 15, 3 (2024), 1553–1566. doi:10.1109/TAFFC.2024.3354382
- [42] Yue Zhang, Chengtao Peng, Qiuli Wang, Dan Song, Kaiyan Li, and S. Kevin Zhou. 2025. Unified Multi-Modal Image Synthesis for Missing Modality Imputation. *IEEE Trans. Med. Imaging* 44, 1 (2025), 4–18. doi:10.1109/TMI.2024.3424785
- [43] Jinming Zhao, Ruichen Li, and Qin Jin. 2021. Missing Modality Imagination Network for Emotion Recognition with Uncertain Missing Modalities. In *Proc.*

- 59th ACL and 11th IJCNLP (Long Papers), Chengqing Zong, Fei Xia, Wenjie Li, and Roberto Navigli (Eds.). 2608–2618. doi:10.18653/v1/2021.acl-long.203
- [44] Zhenjun Zhao. 2024. Balf: Simple and efficient blur aware local feature detector. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*. 3362–3372.
- [45] Yaozong Zheng, Bineng Zhong, Qihua Liang, Guorong Li, Rongrong Ji, and Xianxian Li. 2023. Toward unified token learning for vision-language tracking. *IEEE Trans. Circuits Syst. Video Technol.* 34, 4 (2023), 2125–2135.
- [46] Yaozong Zheng, Bineng Zhong, Qihua Liang, Zhiyi Mo, Shengping Zhang, and Xianxian Li. 2024. Odtrack: Online dense temporal token learning for visual tracking. In *Proc. AAAI*, Vol. 38. 7588–7596.
- [47] Yaozong Zheng, Bineng Zhong, Qihua Liang, Shengping Zhang, Guorong Li, Xianxian Li, and Rongrong Ji. 2025. Towards universal modal tracking with online dense temporal token learning. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2025).
- [48] Yixiao Zhou, Ziyu Zhao, Dongzhou Cheng, Zhiliang Wu, Jie Gui, Yi Yang, Fei Wu, Yu Cheng, and Hehe Fan. 2025. Dropping Experts, Recombining Neurons: Retraining-Free Pruning for Sparse Mixture-of-Experts LLMs. In *Findings of the Association for Computational Linguistics: EMNLP 2025*. 15169–15186.
- [49] Heqing Zou, Meng Shen, Yuchen Hu, Chen Chen, Eng Siong Chng, and Deepu Rajan. 2024. Cross-Modality and Within-Modality Regularization for Audio-Visual Deepfake Detection. In *Proc. ICASSP 2024*. 4900–4904. doi:10.1109/ICASSP48485.2024.10447248

A Discussion and Variants of Global Semantic Consistency (Definition 3)

This section provides a theoretical discussion and practical extensions of **Definition 3**. We clarify how anchor-based distributions offer a robust alignment scheme across incomplete modalities and propose principled variants to improve generalization.

A.1 Why Anchor Distributions Matter

When modalities are missing, direct feature alignment is unreliable since each lies in a distinct sub-space. We thus map $z \in \mathbb{R}^d$ to a shared anchor-distribution simplex via

$$s(z) = \text{softmax}(C^\top z) \in \Delta^{C-1},$$

where $C = [c_1, \dots, c_C]$, and $s(z)_c$ denotes the affinity of z to class c . This yields three benefits.

Modality-agnostic semantic alignment. Projecting onto the anchor simplex cancels modality-specific distortions. For two samples $z^{(m_1)}$ and $z^{(m_2)}$ from different modalities, semantic agreement is ensured whenever

$$\arg \max s(z^{(m_1)}) = \arg \max s(z^{(m_2)}),$$

thus avoiding brittle feature-level matching and enforcing alignment directly at the category level.

Decision-space consistency across modalities. Unlike contrastive or metric losses in *representation space*, anchor distributions encode semantics directly, enabling alignment in the *decision space*. For local distillation (Sec. 4.3) we minimise

$$\mathcal{L}_{\text{align}}^{(i)} = D(s(z_i^{(m)}), s(c_y^{(m)})),$$

with D as KL or cosine, forcing all modalities to converge on a common anchor simplex, guaranteeing coherent cross-modal behaviour without requiring fragile feature-level alignment.

Uncertainty modelling for missing modalities. Anchor distributions provide a natural estimate of prediction confidence through entropy:

$$H(z) = - \sum_{c=1}^C s(z)_c \log s(z)_c.$$

With all modalities present, features cluster near class anchors, yielding sharp low-entropy distributions; as modalities drop, $s(z)$ flattens and $H(z)$ rises, a direct signal of semantic instability. We exploit this in two ways:

(i) **Entropy-regularised anchor selection** (Sec. 4.2). During Top-K mining we prefer confident samples via

$$\text{Score}_{i,c} = \cos(z_i, p_c) - \lambda H(z_i),$$

where the entropy term penalises ambiguous candidates.

(ii) **Uncertainty-guided completion** (Sec. 4.3). Samples with high $H(z_i)$ are routed to an LLM for semantic denoising or completion, ensuring that only genuinely uncertain cases receive auxiliary guidance.

A.2 Variants of Anchor-Based Semantic Modelling

Definition 3 views semantic consistency as an *anchor-induced distribution alignment* problem. To handle heterogeneous or incomplete modalities, we propose three extensions improving *flexibility*, *stability*, and *performance*.

Flexible similarity metrics. The default score $\langle c_k, z \rangle$ may suffer from scale mismatch. Alternatives include:

- **Cosine similarity.**

$$\cos(c_k, z) = \frac{\langle c_k, z \rangle}{\|c_k\| \|z\|},$$

which mitigates feature-norm disparities.

- **Non-linear projection:**

$$s(z) = \text{softmax}\left(\frac{1}{\tau} [f_1(z), \dots, f_C(z)]\right),$$

where learnable heads $f_k(\cdot)$ enhance expressiveness.

Both improve adaptability without losing interpretability.

Adaptive temperature scaling. The sharpness of anchor distributions is controlled by τ . We set it dynamically by sample confidence:

$$\tau_i = \tau_0 + \alpha \cdot H(s(z_i)),$$

so high-entropy (uncertain) cases receive softer distributions, reducing overconfidence and stabilizing training.

Expanded Distribution Divergence Measures. The original constraint in Definition 3 uses L_1 distance between anchor distributions:

$$\|s(z_p) - s(z_q)\|_1 \leq \epsilon.$$

We generalize this to incorporate richer distribution alignment criteria, offering finer control over semantic agreement:

- **KL Divergence (Kullback–Leibler):**

$$D_{\text{KL}}(s(z_p) \| s(z_q)) = \sum_{c=1}^C s(z_p)_c \log \frac{s(z_p)_c}{s(z_q)_c}$$

Emphasizes directional divergence, commonly used in teacher-student distillation.

- **JS Divergence (Jensen–Shannon):**

$$D_{\text{JS}}(s(z_p), s(z_q)) = \frac{1}{2} D_{\text{KL}}(s(z_p) \| m) + \frac{1}{2} D_{\text{KL}}(s(z_q) \| m),$$

where $m = \frac{1}{2}(s(z_p) + s(z_q))$. A symmetric and bounded variant that improves robustness under uncertainty.

Table 5: Hierarchical anchor design as structured instantiations of Definition 3.

Level	Anchor Type	Construction Method	Role
1 (Static)	p_c	Prompt-based LLM encoding	Language prior, semantic reference
2 (Local)	$c_c^{(m)}$	Top- K confident unimodal features	Intra-modal alignment, local supervision
3 (Global)	$c_c^{(g)}$	Top- K structure-enhanced fused embeddings	Inter-modal consensus, distillation target

• **Wasserstein Distance (Earth Mover’s Distance):**

$$D_W(s(z_p), s(z_q)) = \inf_{\gamma \in \Pi(s(z_p), s(z_q))} \mathbb{E}_{(c_1, c_2) \sim \gamma} [\|c_1 - c_2\|]$$

Measures minimal semantic transport to transform $s(z_p)$ into $s(z_q)$, capturing content and structure alignment.

These allow tuning sensitivity to dataset traits and can be staged (e.g., KL early, JS later).

Hierarchical Anchor Construction. While Definition 3 assumes a static anchor matrix C , real multimodal data show rich intra- and inter-modal variation. We thus adopt a three-level anchor hierarchy, each capturing a different granularity of alignment. Table 5 summarizes their construction and roles.

Interpretation. All three anchor types share the same distribution form but differ in abstraction, source, and role. The hierarchy enables coarse-to-fine propagation: static anchors give language priors, local anchors keep modality distinctions, and global anchors unify cross-modal consensus, achieving robust alignment even with partial or ambiguous data.

Derivation of Definition 3

To explore the theory of global semantic consistency (Definition 3), we build on intra-instance (Def. 1) and inter-instance coherence (Def. 2), and present a unified anchor-alignment paradigm.

Unified Modeling Perspective: Definition and Comparison. We define a shared anchor matrix $C = [c_1, \dots, c_C] \in \mathbb{R}^{d \times C}$, where c_k is the prototype of class k . For any $z \in \mathbb{R}^d$,

$$s(z) = \text{softmax}(\tfrac{1}{\tau} C^\top z) \in \Delta^{C-1},$$

with $\tau > 0$. Definition 3 requires same-class pairs to satisfy

$$\|s(z_p) - s(z_q)\|_1 \leq \epsilon, \quad \text{if } y_p = y_q.$$

This differs from Def. 1 (intra-instance, bounding $\|\phi_m(x^{(m)}) - \phi_n(x^{(n)})\|_2$) and Def. 2 (inter-instance, bounding $\|\Psi(\cdot) - \Psi(\cdot)\|_2$). Def. 3 instead aligns all representations in a shared anchor space, yielding more robust and discriminative signals.

Error Analysis: Comparing Three Paradigms. Consider a general similarity objective over pairs:

$$\max \sum_{i,j} Y_{ij} \rho^+(z_i, z_j) + (1 - Y_{ij}) \rho^-(z_i, z_j),$$

where $Y_{ij} = 1$ for positive pairs (same class), and ρ^+ , ρ^- denote similarity measures.

(1) Instance-level Consistency: Cross-modal pairing risks false negatives when unlabeled same-class pairs are treated as negatives, with error $\epsilon_{ic} \propto P(y_i = y_j | I_{ij} = 0) \approx \frac{1}{C} + \beta r$.

(2) Cluster-level Consistency: Constraints after clustering suffer semantic ambiguity and distribution shift, decomposed into δ_{se} and δ_{st} , giving $\delta^{m,n} = \delta_{se}^{m,n} + \delta_{st}^{m,n}$.

(3) Semantic Anchor Alignment (Definition 3): Projecting onto shared anchors reduces both false negatives and cross-modal ambiguity; since $\arg \max s(z) \in \{1, \dots, C\}$ identifies class semantics, error $\epsilon_{sc} \rightarrow 0$ under ideal conditions.

Objective and Optimization Stability. The alignment objective under Definition 3 becomes minimizing distribution divergence:

$$\mathcal{L}_{sc} = \sum_{y_i=y_j} \|s(z_i) - s(z_j)\|_1,$$

with stability from (i) shared anchors enforcing a common space, (ii) independence from explicit cross-modal alignment, and (iii) smooth gradients $\nabla_z s(z) \propto C$ that avoid unstable soft-label fluctuations.

Derivation of (Def. 3 $\Leftarrow$ Defs. 1–2). Assume the *instance-level* and *inter-instance* coherence constraints hold:

$$\|\phi_m(x_i^{(m)}) - \phi_n(x_i^{(n)})\|_2 \leq \epsilon_1, \quad \|\Psi(\{x_i^{(m)}\}) - \Psi(\{x_j^{(m)}\})\|_2 \leq \epsilon_2.$$

for any modalities m, n of the *same* sample and any samples i, j of the *same* class.

Step 1: Bounding representation discrepancy. For any same-class pair of embeddings z_p, z_q —each being either modality-specific $\phi_m(\cdot)$ or fused $\Psi(\cdot)$ —the triangle inequality yields

$$\|z_p - z_q\|_2 \leq \epsilon_1 + \epsilon_2. \quad (1)$$

Step 2: Mapping to anchor distributions. Let $C = [c_1, \dots, c_C]$ be the class-anchor matrix and define

$$s(z) = \text{softmax}(\tfrac{1}{\tau} C^\top z). \quad (2)$$

Because soft-max is 1-Lipschitz under ℓ_2 after scaling by $\frac{1}{\tau} \|C^\top\|_2 = L$,

$$\|s(z_p) - s(z_q)\|_1 \leq L \|z_p - z_q\|_2. \quad (3)$$

Step 3: Combining (1) and (3). Substituting (1) into (3) gives

$$\|s(z_p) - s(z_q)\|_1 \leq L (\epsilon_1 + \epsilon_2) =: \epsilon. \quad (4)$$

Step 4: Consistency in the limit. As $\epsilon_1, \epsilon_2 \rightarrow 0$, the bound $\epsilon \rightarrow 0$, so same-class embeddings converge in the anchor-distribution simplex:

$$\lim_{\epsilon_1, \epsilon_2 \rightarrow 0} \|s(z_p) - s(z_q)\|_1 = 0.$$

Equation (4) is precisely the semantic-consistency condition of Definition 3, showing that Defs. 1–2 together *imply* Def. 3 and thereby unify representation-space and decision-space alignment.

Discussion. Unlike contrastive learning with pairwise sampling, Def. 3 introduces learnable anchors for a stable, discriminative, and modality-robust objective. This reduces false negatives and cross-modal ambiguity, and yields stronger performance under missing modalities and class sparsity.

Derivation of Proposition 1. Let $z_p, z_q \in \mathbb{R}^d$ be embeddings of the same class, either modality-specific $\phi_m(x)$ or fused $\Psi(\cdot)$. With anchor matrix $C = [c_1, \dots, c_C]$ and temperature τ , define

$$s(z) = \text{softmax}(\frac{1}{\tau} C^\top z).$$

Under Defs. 1–2,

$$\|s(z_p) - s(z_q)\|_1 \leq \epsilon, \quad \epsilon \xrightarrow{\epsilon_1, \epsilon_2 \rightarrow 0} 0.$$

PROOF. (1) Bounding representation distance. From Defs. 1–2 we have, for any same-class pair,

$$\|z_p - z_q\|_2 \leq \delta, \quad \delta := \max(\epsilon_1, \epsilon_2). \quad (1)$$

(2) Soft-max Lipschitz property. Set $v_p = \frac{1}{\tau} C^\top z_p$ and $v_q = \frac{1}{\tau} C^\top z_q$. Because the soft-max is 1-Lipschitz w.r.t. ℓ_2 ,

$$\|s(z_p) - s(z_q)\|_1 \leq \|v_p - v_q\|_2 = \frac{1}{\tau} \|C^\top (z_p - z_q)\|_2. \quad (2)$$

(3) Spectral-norm bound. Applying $\|Ax\|_2 \leq \|A\|_2 \|x\|_2$ to (2) yields

$$\|s(z_p) - s(z_q)\|_1 \leq \frac{1}{\tau} \|C^\top\|_2 \|z_p - z_q\|_2 = L \|z_p - z_q\|_2, \quad L := \frac{1}{\tau} \|C^\top\|_2. \quad (3)$$

(4) Combining (1) and (3). Substituting (1) into (3) gives the desired bound

$$\|s(z_p) - s(z_q)\|_1 \leq L \delta = L \max(\epsilon_1, \epsilon_2). \quad (4)$$

Since L is constant (anchors C and τ are fixed), $\epsilon_1, \epsilon_2 \rightarrow 0$ implies $\|s(z_p) - s(z_q)\|_1 \rightarrow 0$, proving that same-class embeddings converge in the anchor-distribution simplex. $\square$

B Additional Implementation Details

For reproducibility we provide (i) the exact LLM prompt templates for semantic anchoring, and (ii) the full training schedules, hyper-parameters, and other settings needed to replicate our results.

B.1 Prompt Templates

To leverage LLMs for multi-modal emotion classification, we design structured prompt templates integrating modality cues and class semantics, yielding discriminative prototypes for distillation.

(1) Prompt Template Structure. Each prompt includes:

- Learnable prefix tokens $P^{(c)} = [P_1^{(c)}, \dots, P_k^{(c)}]$;
- A modality token $[\text{MOD}]$ (e.g., indicating text, video, audio);
- Category Tokens $\text{CLS}^{(c)} = [\text{CLS}_1^{(c)}, \dots, \text{CLS}_S^{(c)}]$.

(2) Prompt Composition. The prompt sequence is:

$$T_p^{(c)} = P^{(c)} \parallel [\text{MOD}] \parallel \text{CLS}^{(c)}$$

combining implicit semantics, modality, and category info. **(3) Training and Optimization.**

- Prefix tokens are randomly initialized and optimized via gradient descent.
- Modality and category tokens are fixed for stable guidance.
- The full prompt is fed into a frozen LLM; output embeddings serve as multi-modal semantic prototypes.

The modality token acts as a bridge, enabling semantically consistent but modality-adapted representations, enhancing robustness to missing modalities and cross-modal transfer.

(4) Examples

Binary Classification:

- Positive (video):

$[P_i^{(Pos)}] \parallel \text{This is a video.} \parallel \text{positive, happy, satisfied}$

- Negative (audio):

$[P_i^{(Neg)}] \parallel \text{This is an audio clip.} \parallel \text{disappointed, sad, angry}$

Multi-class (7-class) Examples:

- Joy (text):

$[P_i^{(Joy)}] \parallel \text{This is a text.} \parallel \text{happy, joyful, content}$

- Anger (video):

$[P_i^{(Anger)}] \parallel \text{This is a video.} \parallel \text{angry, furious, intense}$

- Sadness (audio):

$[P_i^{(Sadness)}] \parallel \text{This is an audio clip.} \parallel \text{sad, upset, down}$

This design enables fine-grained, modality-aware representations that improve classification robustness and support distillation.

B.2 Training Protocol

Missing-modality regimes. We benchmark two settings: (i) *fixed* masks (e.g., $\{L\}$, $\{A, V\}$); (ii) *random* masks dropping each modality independently yet keeping at least one view. The missing rate is

$$r_{\text{miss}} = \left(1 - \frac{1}{NM} \sum_{i=1}^N m_i\right) \times 100\%,$$

with $M = 3$ modalities and m_i the number observed for sample i . We sweep $r_{\text{miss}} \in \{0, 10, \dots, 70\}\%$, where 70% approximates the theoretical limit $(M - 1)/M$.

Hyper-parameters. Table 6 lists all model, optimizer, and loss settings required for exact replication.

Table 6: Full training and configuration details.

Component	Value
Text encoder	BERT-base (768 d)
Visual encoder	Facet (35 d)
Acoustic encoder	COVAREP (74 d)
Shared embedding	512 d
Prompt encoder	Frozen T5-Base
Optimizer	AdamW
LR (BERT / others)	$5 \times 10^{-5} / 1 \times 10^{-3}$
Batch size / epochs	32 / 20
Graph k, δ, τ, β	15, 1.5, 0.7, 0.5
Loss weights ($\lambda_{\text{loc}}, \lambda_{\text{glob}}, \lambda_{\text{p}}$)	(1.0, 1.0, 0.5)
Anchor confidence γ	0.3

Each experiment was repeated five times with different random seeds, and the mean of the test-set metrics is reported. All training and evaluation were conducted using PyTorch with CUDA on a single NVIDIA A800 GPU.